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1 import time
2
3
4 students = []
5 # --- 1. Storage (The Data) ---
6 # This list holds all the student records you enter
7
8
9 # --- 2. Functional Requirements (The Logic) ---
10 def add_student():
11     global students
12     """Function to input and process a student
    record"""
13     print("\n--- Add a New Student ---")
14     name = input("Enter student name: ")
15     student_id = input("Enter student id: ")
16     grade = input("Enter Student grade (0-100): ")
17     # Create a record (Dictionary)
18     record = {"name": name,
19               "student_id": student_id,
20               "grade": grade}
21     # Process: Add the record to our list
22     students.append(record)
23 print("Record saved successfully!")
24
25
26 def view_records():
27     global students
28     """Function to display the processed results"""
29     print("\n--- Current Student Records ---")
30     if not students:
31         print("No records found. please add a student first
    .")
32     else:
33         # Loop through each record to display them
34         for s in students:
35             print(f" ID: {s['student_id']} | Name: {s['name']}
    | Grade: {s['grade']}")
36         print("\nReturning to menu in 5 seconds...")
37         time.sleep(5)
38

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39
40  # --- 3. The Main Terminal Interface (The Loop) ---
41  def main():
42      while True:
43          print("\n=====")
44          print("STUDENT MANAGEMENT SYSTEM")
45          print("=====")
46          print("1. Add Student Record")
47          print("2. View All Records")
48          print("3. Exit Program")
49          choice = input("\nSelected an option (1-3): ")
50      )
51      # Decision structures (if, elif, else)
52      if choice == '1':
53          add_student()
54      elif choice == '2':
55          view_records()
56      elif choice == '3':
57          print("Closing system. Goodbye!")
58          break # This stops the loop and exits
59      else:
60          print("Invalid input! please type 1' 2, or 3
61          .")
62      # This starts the application
63  main()
```